FOUNDATION COURSE

START YOUR
PREPARATION
FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 21

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Recap *of previous lecture*

INTERPOLATION



Topics *to be covered*

" PERMUTATION-COMBINATION "



Counting Principle

① f.P of Addition ⟶ ? Either-or / only one.

② fP of Multiplication ⟶ ? And / All

(*) If we have to perform only one of the job (out of n jobs) at a time then use Addition P.

(*) " " " All the jobs (" ") " " " Multi P.

Eg In a Restaurant there are (8) Veg dishes & (5) Non Veg dishes then in how many ways you can order a dish?

Total ways $= 8 + 5 = 13$

Here we can order either Veg dish (or) Non Veg dish

Eg If there are 15 NITS & 20 IITS in INDIA & you are selected in JEE then in how many ways student can choose a College?

Sol: Either student can select NITS (or) he/she can select IITS

Total ways $= 15 + 20 = 35$ ways.

Eg: If there are 20 IITs, each having 7 Branches then in how many ways toppers can take admissions?

Total ways = (selection of IIT) & (selection of Branch)

= 20 × 7 = 140 ways.

11, 21, 31
12, 22, 32
13, 23
14, 23
15, 24
16, 28
17, 28, 3) - - - 3(20,7)

Eg: There are (8) Boys & (5) Girls in a class then in how many ways teacher can select

(1) either a Boy (or) a Girl = ? = 8 + 5

(2) A Boy & A Girl = ? = 8 × 5

Eg: How many 3 letter words can be formed (with/w/o meaning) using Eng. alphabets. (26)

① if Repition is Not allowed = ? $= \underset{P_1}{26} \times \underset{P_2}{25} \times \underset{P_3}{24} = {}^{26}P_3$

② if " is allowed = ? $= \underset{P_1}{26} \times \underset{P_2}{26} \times \underset{P_3}{26}$

Eg How many 4 digit Nos can be formed using 1, 3, 5, 7, 9

① RNA = ? $= \underset{P_1}{5} \times \underset{P_2}{4} \times \underset{P_3}{3} \times \underset{P_4}{2} = 120$ choices $= 120$ NOS $= {}^5P_4$

② RA = ? $= \underset{P_1}{5} \times \underset{P_2}{5} \times \underset{P_3}{5} \times \underset{P_4}{5} = 5^4$

(P_1: 1,3,5,7,9) (P_2: 1,3,5,7,9) (P_3: 1,3,5,7,9) (P_4: 1,3,5,7,9)

eg: How many (4 digit Nos) can be formed with distinct digits?

four places

RNA

W.K. that we have 10 special digits for eg 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Total (four) digit Nos (RNA) = $\underset{P_1}{9} \times \underset{P_2}{9} \times \underset{P_3}{8} \times \underset{P_4}{7}$

eg 2310 ✓
2415 ✓
5103 ✓
0142 — X
...
?

(ii) If (RA) then Total four digit Nos = ?

$= \underset{P_1}{9} \times \underset{P_2}{10} \times \underset{P_3}{10} \times \underset{P_4}{10}$

Factorial → $\boxed{n! = n.(n-1).(n-2)\ldots\ldots.3.2.1}$; $n \in N$.

eg $5! = 5\times4\times3\times2\times1$, $4! = 4\times3\times2\times1$, $7! = 7\times6\times5\times4\times3\times2\times1$ etc

Sp Case :- $\boxed{0! = 1}$, $\lfloor n$ = Sunsclem if $n \in \bar{I}$ or fraction

eg: How many (5 digit Nos) can be formed using odd digits? (of RNA)

(1, 3, 5, 7, 9)

Total 5 digit Nos (RNA) $= \underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 5!$
$\quad\quad P_1 \quad P_2 \quad P_3 \quad P_4 \quad P_5$

eg How many (4 letter words) can be form using the letters "ROSE" if each letter is coming exactly once ?

RNA

Total words $= \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 4!$
$\quad\quad P_1 \quad P_2 \quad P_3 \quad P_4$

Q In how many ways 6 persons can be seated on 6 chairs ?

Total seating arrangements = $\underset{P_1}{6} \times \underset{P_2}{5} \times \underset{P_3}{4} \times \underset{P_4}{3} \times \underset{P_5}{2} \times \underset{P_6}{1} = 6! = {}^6P_6$

(RNA)

Q In how many ways we can arrange the letters of the word "EQUATION" if

A, E, I, O, U, Q, T, N

① there is No Restriction = ? $= \underset{P_1}{8} \times \underset{P_2}{7} \times \underset{P_3}{6} \times \underset{P_4}{5} \times \underset{P_5}{4} \times \underset{P_6}{3} \times \underset{P_7}{2} \times \underset{P_8}{1} = 8! = {}^8P_8$

(RNA)

② if words are starting with E = ? $= \underline{1} \times \underline{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} = 1 \times 7!$

③ words starts with E & ends with N = ? $= \underline{1} \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 \times \underline{1} = 1 \times 6! \times 1$

Q How many (4 letter words (W or W/O meaning) can be formed using the letters of the word 'EQUATION' = ?

1 2 3 4 5 6 7 8

RNIA

$$= \frac{8}{P_1} \times \frac{7}{P_2} \times \frac{6}{P_3} \times \frac{5}{P_4} = {}^8P_4$$

Q How many (5 letter words) can be formed using the letters of the word 'LOGARITHMS' ?

1 2 3 4 5 6 7 8 9 10

RNIA

$$= \frac{10}{P_1} \times \frac{9}{P_2} \times \frac{8}{P_3} \times \frac{7}{P_4} \times \frac{6}{P_5} = {}^{10}P_5$$

(1) if $n > r$ & RNA then Multi Rule = Perm Rule

(2) if $n = r$ & RNA then Multi Rule = Perm Rule = Factorial Rule

(3) if RA then only use Multi Rule

(4) Combination → (if only selection is Req)

$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

$^{11}C_3 = \frac{11\times10\times9}{3\times2\times1}$ & $^{22}C_4 = \frac{22\times21\times20\times19}{4\times3\times2\times1}$

(5) Permutation (Selection & Arrangement)

$$^{n}P_{r} = \frac{n!}{(n-r)!} = {}^{n}C_{r} \times r!$$

& $^{11}P_3 = 11\times10\times9$, & $^{22}P_4 = 22\times21\times20\times19$

(X) $^{n}C_{r} = {}^{n}C_{n-r}$ ⇒ $^{25}C_{22} = ? = {}^{25}C_3 = \frac{25\times24\times23}{3\times2\times1}$

Eg In how many ways cricket team can be selected from Batch of 15 players if

(1) there is No Restrictions = ? $= {}^{15}C_{11} = {}^{15}C_{15-11} = {}^{15}C_4 = \frac{15\times14\times13\times12}{4\times3\times2\times1}$

(2) A particular player is always selected = ? $= {}^{1}C_1 \times {}^{14}C_{10} = {}^{14}C_4 = \frac{14\times13\times12\times11}{4\times3\times2\times1}$

(3) " " " is never selected = ? $= {}^{14}C_{11} = {}^{14}C_3 = \frac{14\times13\times12}{3\times2\times1}$

Eg In a Batch of (10) Batsman, (8) bowlers, (5) All rounders & (2) wicket keepers a Cricket team has to be selected in which there must be (5) Batsman, (3) Bowlers, (2) all rounders and (1) W.K then in how many ways this can be done ?

Total ways $= {}^{10}C_5 \times {}^{8}C_3 \times {}^{5}C_2 \times {}^{2}C_1$

Q. How many four letters words can be formed using the letters of the word "FAILURE" if (1) there is no Restriction = ? $= {}^7C_4 \times 4! = {}^7P_4$

1 2 3 4 5 6 7

(2) F is included in each word = ? $= {}^6C_3 \times 4!$ eg A, U, R, F

(3) F is not included in any word = ? $= {}^6C_4 \times 4!$ eg A, U, R, E

Q. out of 7 Consonants and 5 Vowels, how many 5 letter words can be formed including 3 Consonants & 2 Vowels? $= {}^7C_3 \times {}^5C_2 \times 5!$

H, P, D, A, I

① Total Cards (52)

- 26 Black
  - 13 Club
  - 13 Spade
- 26 Red
  - 13 Heart
  - 13 Diamond



② Face Cards (K, Q, J) ⟶ 12 Cards

③ Honour :: (K, Q, J, Ace) ⟶ 16 Cards

Q In how many ways we can select 4 cards out of 52 cards if

(1) there is No Restriction = ? $= {}^{52}C_4$

(2) four cards are of same suits = either (all are spade) or (club) or (Diamond) or (Heart) $= {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4 + {}^{13}C_4$

(3) " " " " different suits = (1 spade) & (1 club) & (1 Diamond) & (1 Heart) $= {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1 = ({}^{13}C_1)^4$

(4) " " " Face Cards $= {}^{12}C_4$

(5) " " " Honour Cards $= {}^{16}C_4$

(6) Two are Black & Two are Red $= {}^{26}C_2 \times {}^{26}C_2$

(7) 4 Cards are of Same Colour = Either (all 4 Black) or (all 4 Red) $= {}^{26}C_4 + {}^{26}C_4$

(*) Prob = $\frac{\text{fav. No. of items}}{\text{Total No. of items}} = \frac{\text{Fav No. of Cases}}{\text{Total No of Cases}}$

Total Case → No Restriction

fav Cases → which is Required

(*) Counting Based on Alike items → eg (1,2,3) ⟹ Total 3 digit Nos = $3! = 6$

& Various Nos are (123), (132), (213), (231), (312), (321)

eg 5,5,6 then Total 3 digit Nos = ? = $\frac{3!}{2!} = 3$ Nos

Various Nos (556), (565), (655)

Eg: 4 letter words using a, b, c, d = ? (different) $= \frac{4}{P_1} \times \frac{3}{P_2} \times \frac{2}{P_3} \times \frac{1}{P_4} = 4! = 24$

Eg 4 letter words using a, a, b, b = ? $= \frac{4!}{2!\,2!} = 6$ Nos

Various Nos are (a a b b), (b b a a)
(a b a b), (b a b a)
(a b b a), (b a a b)

Q. Five digit Nos using 4, 4, 4, 7, 7 = ?
$= \frac{5!}{3!\,2!} = \frac{120}{6 \times 2} = 10$

Eg: How many "11 letter words can be formed using M I S S I S S I P P I"

S S S S I I I I P P M
1 2 3 4 5 6 7 8 9 10 11

$= \frac{11!}{4!\,4!\,2!}$ Ans


Thank
YOU